VU AI HUB · LECTURE NOTES

History of AI: lectures

Year 1, April to May 2026

Written by Inaaya and shared on Notion. Original page: cheddar-wall-f3b.notion.site/3339801a930980d688ddf9192dd79f79

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30th of March

Lecture 1: Introduction

Own notes:

History of Science

Prehistory of Computing

Charles Babbage (1791-1871)

- Inventor of the calculator
- Brilliant mathematician
- Interested in companies
  - Looked into the banks of London, made it more efficient by creating bank accounts

Frederik Taylor (1856 - 1915)

???

- Automation is not new, it has three traditions:
  1. Administration
     - Build on bureaucracies so they can make better decisions
     - Using punch card machines (Hollerith machine) to process data (in the beginning it was counted by humans to check if it was right, but when it checked out they kept using the machine)
  2. Process Control
     - They needed their machines to react like humans
  3. Science and Engineering
     - Computers were a job mostly done by woman
     - Calculations regarding weather prediction calculations, mechanical calculations, aeronautics, econometrics, telephone networks and military applications etc.

Human thinking (and behavior)

Someone from the gc's notes:

Charles Babbage Proposed to the government to speed up.

Success: digging into companies, what they were doing. The banks of London was his main project.

The clearing house is what they called were they collect checks. They exchanged checked. They observed this and suggested to make it even more sufficient. Everyone can take one account in the bank and everyone can do it on paper without going to the clearing house.

Intermezzo

Howard Aiken: he was building one of the first mechanic electrical machine.

Three traditions or three storyline

Administration: In the US, every ten years they were counting there entire population. They needed that data to calculate how many taxes they were going to receive, e.g. what can we spend etc. they went door by door to ask the questions e.g. how many people, what are they aiming for their.. etc. by 1890 the number of data that they collected was huge that they simply needed Hollerith machines to store the data.

They came up with the Hollerith machines. They did it within a year. They did do the counting manual as well to check because this was the first time.

The office in 1920 looked at it, thought it was cool and realizes that it could speed up the time so they used it also.

Calculators and typewriters was also a mechanical that they used. Punched card typing, sorting machines, counting machines, tabulating machines. IBM was the main of that.

They started efficiently movement. They were selling these machines to big companies, governments etc. It wasn't necessary all very high tech. libraries would use simple cards to organize everything. It was these kind of machines that administration worked.

All these administrations had these outlines/flowcharts of what was going where. That was the main idea.

Process control:

This was the closest as of what has become AI. All these companies were using these e.g. Shell/ Hogevens.

They needed things in checked so they used outlines analogue to the process. It's partly electrical partly mechanical. Tyring to place human decision. That was the main idea. They also had the outlines. indication what and when and how decisions were made.

Science and engineering:

Closest to university. It has computer actives. They were high educated women that used those computers. They connected calculations to the decide the outcome. All sort of calculations were made by hand or mechanical calculations. D. Hartree came up with numerical Analysis. Schematic and flowcharts how would you do those high calculations. V. Bush came up with differential analyser.

These three traditions have outlines in common. Money is mostly available in administration. They loved that efficiency movement, that's also a place where you would see computers rising after WW2.

A prehistory of AI:

In 1950 when AI rose, that was not from scratch. They had already been familiar with Ai. Two lines is human thinking/behaviour and the second was language.

Human behaviour: do I think therefore I am. This was one start. From Rene Descartes. Everything has it course and the world does not do something different when I observe it. It always does the same. The universal law. In the 70 century they started by mimicking human behaviour and thinking. The same thing with these process control devices. We have them machines that mimics human dependence.

Rise of teaching and school systems brings in more thinking about thinking. How do you get equations in your head, or do you understand them when you have them in your head. What do you need to do as a teacher to do this. One teacher William Jevons tries to mimic the logical part of learning by having a piano. He conceptualized it as a thinking machine.

The 20th century. Chess endgame. Produces completely mechanical. To catch parts of human reasoning to recognize them to automate them, looking if they could reproduce them in any way.

With hindsight: in literature and popular culture, artificially created humans pop up long before suggestions of automatic computers.

No one called computers AI. AI in our sense of the world doesn't exist without computers. But it was already present in the 19th century.

Language is the second thing that's important in AI. Language and human thinking starts with Boolean logic. He presents his books as a investigations of the laws of thought. Expressing in Boolean logic how we think in True and False. You can do anything with them. It was calculation with thoughts/language.

Decision problem: can I always calculate the truth or not truth of a given expression in a given mathematical system and Kurt Gödel says no.

Alonzo Church produces the lambda calculus. The most used language in AI. It always you to calculate with al kinds of logical expressions and come up with conclusions.

Alan turing produced the On computable numbers. it's a conceptual machine that helps him decide given what you can compute.

Learning a language starts with phonology, morphology, syntax, vocabulary. And this is the typical book you get. How to build up and teach the language.

Volapuk is one of the international Artificial language. They wanted a language were they could easily communicate. Constructing a simple and in-between language could help to learn a new language.

Music learning: klavarscribo (1931). Until the 1960 It was widely used by piano players.

8th of April

Lecture 2: Creating Computers

Aditi's notes:

Cold War Science

- Rocket engineers were more popular during the mid forties compared to computer engineers.
- Marian Adam Rejewski (1905-1980), Jerzy Witold Rozycki. And Hennry Zygalski these ppl cracked the enigma code.
- The first computer that consisted of a keyboard was created in Japan, the Americans were not as devastated by the war but wanted to start creating the computer. - Yamashita type statistical Machine (keyboard)
- Computers were efficients but were not really good for the businesses. Computers were not invented by computer scientists and were mainly created by the physicist and mathematicians.
  The fact that was spread across during the time that the computer was very helpful and makes life efficient was a prejudice and was ill informed.
- From the american point of view the cold war science was
- IBM started by building machines for the war
- After WW2 NASA was created

Apart US England was the only country that were able to start creating computers and technologies

- Some continents in Europe started rebuilding the country during the 1940s
- French opposed the American culture. France wanted the technologies and the money but they did not want the american culture.

Cold War Science

- Manchester, december 1948 - the " Manchester baby "

The Dutch Dinosaur

- A room full of human computers calculating the land reclamation and the water works were undertaken during the 1910s and the 1920s by H.A. Lorentz and C.Lely.
- These were analogs: the tradition of process control: engineer Johan van veen (1893- 1953).
- They built the Deltar, an analog machine which calculated the water flows and was used until 1972. Rijkswaterstaat another analogy machine that was used.
- 1946 directly after the war the government started rebuilding the country all the money was sent to amsterdam and these money were given to the mathematicians and were asked to create the (calculating thing regarding water and airlines)- Mathematish (1946) - Moore School - Started creating centrum and people started doing calculation to rebuild the netherlands
- Aad Van Wijngaarden (1916 - 1987) , from 1947 onwards director of the computing department of the mathematisch centrum.
- ARRA (computer) was created and was first tested on June 21, 1952.
- A computer that would generate a random number and produce tables of squares. (only worked once and it did not work again, it just generated on random number and then did not work) The first one was a flop and they bluffed their way into creating a better machine and they built the ARRA 2 in march 1954.

The Netherlands needed the technology mainly to build the country so the dams and it was very important in the calculations so that the country is not in flood and devastation. (process control)

- ARRA 2 was put to production. (still human computers had more job even after the production of ARRA 2) - Human computers were still imp until 1960. And 1961 there were more computer machines.
- However The NASA calculation trusted the human computer more for the spatial calculations.
- European Dwares
- Electrologica (1956) - Dutch dwarf
- Zeus KG (1949) - german dwarf
- Regnecentralen (1955) - Danish dwarf was one of the people who helped create computers in denmark.

Delft

- facilitating the telephone traffic (ptt): Zero (), 1951. Ptera
- beginning 1955 ZEBRA (Zeer Eenvoudig Binair)

Eindhoven

- PETER (Philips Elementair Tweetalig Elektronisch Rekenapparaat). Philips had a gentlemen's agreement that Philips would not enter the computer industry as IBM was scared and IBM agreed that they would buy the hardware parts from Philips. After a good amount of years Philips entered the computer industry.
- Shell 1953 made the newspaper with their machines: MiRACLE (Mokums Industrial Research Automatic Calculator for the laboratory and Engineering)

Artificial Intelligence

- Turning 1950 - Dartmouth 1956 Various strands of research already exist. Research field - Cybernetics.
- During the 1950s AI did not exist but Cybernetics existed and AI started during the 1960s
- A.M Turning, intelligent machinery 1948; computing machinery and intelligence 1950
    - Not research disciplines, but psychologists artists and other field

- Cybernetics
- Psychologists
- Language
- Mathematics - formal logics and proof checking
- Philosophy
- Biologists - animal behaviour evolutionary and population

The Dartmouth summer research project on AI

(cambridge department of language was not a part of cambridge university)

1959 conference the calculating machines and human thinking - january 1951.

Tortoise 1948 was used to present an electronic version of animal

13th of April

Lecture 3: The sound of software

Transcription Notes:

Computer Sounds and Auditory Monitoring

- Early computers used sounds as a primary way for operators to monitor program execution and identify problems
- Wilkes explained in the first EDSAC handbook how to use machine sounds to verify correct program operation
- If a computer went silent, it indicated no program was running - machines should click, hum, or make some sound to show they were computing
- Christopher Strachey sent a punched card tape to EDSAC to execute code that would make the computer hum "God Save the Queen" as a test of understanding how computers worked
- ARRA II (1954) used transistors and included speakers to interpret what was happening, similar to earlier computer generations
- Multiple reasons for computer sounds:
  - Announcing program termination
  - Auditory monitoring of execution
  - Specific navigation within a program (mathematicians could tell where the program was at)
  - Signaling program location/status
- Most computers from the 1950s-60s had speakers built into control panels, often with no off button - operators could only muffle them with pillows
- The "beep" command became very popular among scientists for signaling
- Soviet computers like the Zuse Z22 (1957) also featured speakers prominently
- Later in the 1960s, computers began being used for actual music (not just beeps), with Jean-Michel Jarre becoming world famous for computerized music

Use of Metaphor in Computing

- Metaphors made it possible to communicate about new computing concepts to outsiders and provided reassuring ways to address unfamiliar technology
- **"Program"** originated from theater/radio programs - a sequence of events - which engineers adapted for computing
  - First appeared in 1946 with ENIAC, where women were "preparing programs"

    - "Programming" as a verb emerged in 1948
    - The metaphor evolved from a noun describing a sequence to an active verb
- **"Memory"** was one of the first and most significant metaphors
    - Von Neumann already used it in the EDVAC description
    - Engineer Eckert disliked the biological metaphor because he felt it didn't do justice to the engineering work involved
    - Despite resistance, the memory metaphor became widely accepted and was used for mercury delay lines, storage, and capacitors
- **Compiler** originally meant physically stacking punch cards on top of each other, then evolved to mean translating code
- **Assembly** referred to assembling one pile of punch cards with another
- Other terms used: auto-coding (short for automatic coding), sequencing commands
- The Friden STW10 calculator (popular in the 1950s-60s) used "program" to refer to machine settings for multiplication

Metaphors Firing Back: Computing Influences on Human Understanding

- The computer became a "defining technology" that influenced how people understood themselves
- People began investigating human thinking through the perspective of computers - the metaphor "fired back"
- Terms like "electronic brain" and "mechanical brain" became common in the 1950s
- Alan Turing's "Intelligent Machinery" (1948) raised questions about whether machines can think and what thinking actually means
- Karl Lashley worked in the 1920s-30s on measuring human intelligence and simulating brain functions (psychometrics)
- William Grey Walter published *The Living Brain* (1953), discussing the memory metaphor and describing how simple machines could mimic complex behavior
- Cybernetics aimed to emulate human brain functions, either to learn from them or make use of them
- Grace Hopper called programming "educating a machine" (didn't stick as a metaphor)
- Edsger Dijkstra called programming "communication with an automatic computer"
- The question of when machine thinking became distinguished from human thinking remains debated - some argued human thinking is different because it includes history

Real-Time Computing

- Real-time computing was pushed by military agendas in the 1950s (Project Whirlwind, SAGE defense system) and market economics (SABRE)
- Originally, computers were batch job machines - you couldn't observe processing, only listen to it
- SAGE (Semi-Automatic Ground Environment) was the most expensive program that "ever failed" - became operational just as intercontinental ballistic missiles made it obsolete
- IBM, Burroughs, and National Cash Register were the three main companies involved in creating real-time systems
- Computer scientists like Dijkstra initially resisted real-time computing, arguing it spoiled the logical processes they had developed

Barclays Bank and Burroughs Case Study

- Barclays was one of the first banks in England and Europe to adopt computer technology
- They used "robot cashiers" where customers inserted punch cards to withdraw cash

- Looking ahead to D-Day (Decimal Day, February 15, 1971), Barclays needed to convert their systems from pounds/shillings/pence to decimal currency
- Burroughs promised to build the B8500, a larger real-time machine specifically for Barclays
- In 1968, a Barclays programmer discovered Burroughs had abandoned the B8500 project as "too big"
- When testing began on the B6500 (smaller version), electromagnetic interference from a nearby railway station caused the machines to malfunction
- In 1969, production of the B8500 was cancelled entirely
- The final result: Barclays got a system that worked within "a couple of hours" rather than real-time, but still faster than before
- Later in the 1970s, Barclays teamed up with IBM to achieve real-time connections
- This case exemplified both market push (Burroughs) and market pull (Barclays)

Cybernetics in Business

- In 1956, a conference in Namur, Belgium (the birthplace of AI) founded the International Association for Cybernetics
- Notable attendees included Grey Walter, Stafford Beer, and French engineers - but American cybernetics leaders like Wiener and McCarthy didn't attend
- Two separate branches of cybernetics emerged: US and British, which didn't attend each other's conferences
- The 1957 conference discussed Rosenblatt's Perceptron before its 1958 publication - this was the first machine learning attempt using neural networks to distinguish shapes

Stafford Beer and Systems Thinking

- Stafford Beer connected operations research to cybernetics and business to cybernetics in 1956
- He was among the first (in 1956) to idealize having display screens to follow business flow, preferably in real-time
- In 1972, Beer developed the Viable Systems Model with five subsystems to describe company operations
- Similar to universal product codes for inventory tracking - real-time systems to monitor business flows
- Robert Widener created a "management executive armchair control panel" in 1968 to follow business flows in real-time
- Jay Forrester built "World Dynamics" (1971), modeling population growth and resource systems, which became the groundwork for the Club of Rome report (1972)

Project Cybersyn

- In 1971, Chilean President Salvador Allende commissioned Stafford Beer to create Project Cybersyn
- Cybersyn was a viable systems model where every factory in the country would have a system to monitor and control production
- The project included three layers of automation and decision-making, all connected in real-time
- The project was never fully implemented due to the 1973 coup
- Jon Agar's book *The Government Machine* argues that computing success fitted bureaucracy's blueprint - following rules and enabling centralized control
- Cybersyn represented "the computer ruling Chile" - ideas about automated governance

1984 by George Orwell

- George Orwell's 1984 (published in 1949) was widely read by everyone in early cybernetics and computing science communities

- The book describes a dystopian world with three superpowers continually at war, using conflict to maintain social control
- The protagonist works for the Ministry of Truth, continually rewriting history to match current political narratives
- Key themes relevant to computing/AI: totalitarianism, surveillance society, censorship, and thinking in systems
- The concept of "Big Brother" became a reference point for warnings about computers taking over or surveillance
- The book contains no mention of computers or AI - only television screens that work both ways
- Core message: "Control the past, control the future; control the present, control the past"
- The book was mind-blowing for the cybernetics community and influenced how they thought about the societal implications of their work

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Own notes:

IBM outpaced all of their competitors

Christopher Strachey send a punched card tape suggesting to execute the code on EDSAC: humming 'God save the Queen'. They did this to see if they understood how the computers worked.

ARRA II (1954): transistors, with speakers to be able to interpret what was going on like with the computers of the previous generations

All kinds of other speakers

Sounds the computer made showed it was 'thinking' or computing

Sounds were used for various reasons:

- Announcing the program had terminated
- Auditive monitoring
- Specific navigation within a program (mathematicians knew where the program was at)
- Part of the instruction (program is now here)

Use of metaphor

Using metaphors made it possible to speak to people on the outside

ENIAC were preparing programs

- Programming as a verb became a verb in 1948

Friden STW10 was a fully automatic calculator

- It used 2 keyboards to type 2 numbers or decimals to multiply with one another

In 1960 the programming language compiler (stacking on top of each other) became a translator

'Memory' is one of the first metaphors that popped up. Designer didn't like the biological metaphor because he thought it didn't do justice to all the work he put in that thing for it to be called memory.

The computer didn't only start using words and integrating it into their language, but it happened the other way around as well, people started investigating human thinking through the perspective of the computers.

Electronic brain/mechanic brain and Turing's Intelligent machinery (1948) raises the question if these machines can think and what it actually means to think.

Karl Lasley was trying to measure human intelligence, he was trying to simulate brain functions (psychometrics)

William Grey Walter published *The living brain* and talked about the metaphor 'memory' and describes metaphors.

Cybernetics aim to emulate the human brain (or functions that 'belong' to the brain in our perception), either to learn from it or to make use of it.

- Grace Hopper called programming 'educating a machine', but it didn't stick
- Edsger Dijkstra called programming 'communication with an automatic computer'

Real time computing

Realtime computing was really pushed by agendas (like the military in the 1950s, who could use real time responses from machines: project whirlwind, SAGE defense system) as well as down-to-earth economics (for the market push: SABRE)

Barclays was one of the first banks in England, even Europe, to adopt computer technology. They used a robot cashier, where the people came with a punch card, put it in the machine, and the machine gave them the cash. In 1965 they looked toward D-Day, which was an estimated time when pounds would become decimal, this happened on February 15, 1971. They had to change their machines to be able to handle the decimals that were to come before 1971.

Burroughs build the SAGE computer in 1965 which started in banking.

They wanted to make a bigger machine and changed B825 to B8500, they only had the idea but they kinda gave up on it. The Barclays invested in it and wanted to see results, so Burroughs try to make the machine. When they started testing the computer, the communication between the machines was fine. The closeness of a railway station was the cause for the machines not working as they were supposed to. In 1969 the production of the B800 was canceled, which looked bad for both companies because they already advertised the product. They ended up bringing something out, but it couldn't do it in real time, it took a few hours, which was still faster than before. The Barclays teamed up with IBM to get real time connections.

Cybernetics in business

Conference in 1956 (which was the birthyear of AI) was attended by leading computer scientists of that time. This is where the International Association for Cybernetics was founded. Other prominent characters in this world didn't attend, they were mostly Americans. There were two branches of Cybernetics, the one from the US and the Britain one, they didn't attend the conferences from one another.

Stafford Beer connected operations research to cybernetics, and business to cybernetics in 1956. He tried to develop screens so that they could display the computers outputs and in general developing ways business could use computers intelligently. However, he was not the only one doing this, W. Robert Widener did this as well. Jay Forrester built a system 'World dynamics' in 1971 modelling the population growth and other things. Stafford Beer was not only focusing on business, project Cybersyn in 1971 was a viable systems model, where every factory in the country would have a system that would keep everything in check.

1984 - George Orwell

ask Gemini the connection between this book and the History of AI

20th of April

Lecture 4: Computing crisis?

Transcription Notes:

Early Computing and Programming Methods

- Early computers like the X8 machine used console switches for programming, featuring 27 lights and switches for setting registers and addresses
- Programming without punch cards or punch tape required manually setting switches, which was an extremely painstaking process
- Connecting cables to create different paths on the machine was an alternative method, though debugging was difficult

Development of Programming Languages and Compilers

- Early compilers were not efficient, and human-generated code was often more optimal than compiler-generated code
- Programming languages emerged because humans cannot easily understand machine code
- Different programming paradigms developed driven by specific communities (AI, logic, etc.)

ALGOL and the Universal Language Movement

- ALGOL 58 and ALGOL 60 were developed as attempts at a universal programming language
- The language was embraced by the ACM and designed with the idea it should work with a global compiler
- Multiple groups were involved: American Association for Electrical Engineers, IFIP (International Federation of Innovation Professionals)
- Eastern European countries and mathematicians strongly supported the idea of a universal mathematical language
- The ALGOL effort started in Germany with mathematicians and professors discussing language structure
- By 1960 (ALGOL 60), the language was renamed but allowed for additions and modifications
- The universal language ideal eventually fractured as groups split up over disagreements

LISP and Alternative Paradigms

- LISP was based on ideas of recursion and testing, developed within the AI community
- LISP used a distinct syntax where functions and variables were nested (e.g., plus a b)
- Different languages were tied to specific communities and their goals

The Software Crisis

- In the 1960s, the software crisis emerged as programs became too large and complex
- The OS/360 project at IBM was a major example of large operating systems going wrong
- The crisis was particularly acute in the US with companies facing expensive problems

Software Engineering as a Discipline

- Software engineering emerged as a response to the crisis around 1968
- The approach emphasized proving programs step by step to ensure correctness
- This contrasted sharply with the chaotic situation in many US companies

Mathematics and Formal Verification

- Dijkstra advocated for a close relationship between computation and mathematical logic, comparing it to the relationship between analysis and physics in the previous century
- Early proof checkers were developed, including efforts in the Netherlands by De Bruijn around 1967
- The four-color theorem proof by Haken and Appel was controversial because it relied on computer verification rather than traditional mathematical proof
- AURA (Automatic reasoning system) was developed at Argonne National Laboratory in Chicago around 1976
- There was significant influence between mathematics and programming, with proof checkers requiring users to learn specialized languages

Computer Education Reform

- In the 1960s, educational systems across countries were being redesigned to provide universal access to secondary education
- A 1961 British government committee examined what should be included in the new curriculum, publishing their report in 1968
- The report "Computer Mathematics" recommended including programming in secondary education to prevent technological illiteracy
- The committee emphasized that students should understand that humans decide in advance how computers respond, not the computers themselves
- By 1969, proposals called for pupils to use actual computers with simple programming languages, despite most schools having no access to computers
- Solutions included using computer centers where students could send punch cards by mail and receive results
- Some schools used expensive dial-in telephone connections to computer centers
- Programming education aimed to teach problem-solving skills and step-by-step thinking

Educational Psychology and Programmed Instruction

- New educational approaches in the 1960s emphasized understanding rather than mere imitation
- Programmed instruction methods broke down problems step-by-step, allowing students to check their work progressively
- These methods used textbooks with answers on the right side of pages, encouraging students to work through problems independently
- The approach was influenced by computing ideas and allowed for individualizing education

Computer-Assisted Instruction and Logo

- Project PLATO at Illinois University was an early example of computer-assisted instruction on a time-sharing system
- Logo with turtle graphics was developed to teach programming to children, using simple commands to control a drawing robot
- However, computer-assisted instruction never became a serious core part of education, remaining primarily a tool for teacher trainers

Nahid's notes:

computing crisis

- alternatives in programming within whirlwind, people were thinking of setting up the program in a more efficient way.
- auto-coding
- fortran 1957 and cobol 1959 → program started from flowchart
- this wasn't very efficient at first because is was slow, but it got better over time. it was a risk people were willing to take.

programming

- programming language ALGOL 60 → universal which is ideal for both machine and language. It was elegant for EU.
- for US it was more an academic way, not flexible and difficult to learn. they wanted fortran

IAL

- German in origin → frustration because it was not living up to what it was supposed to become.

John McCarthy, LISP

- based on the idea of recursion and infinite nesting
- became the most used language in AI community

what was the problem

- cost of new machine and code
- making existing program run on a new machine, software transfer mostly not possible. computer companies had ready to run programs/libraries.
- for mathematicians a lot of projects were driven by them, used as a challenge of how well a computer can understand logic.

who had agenda's

- ACM- the reason: they wanted to be able to explain how it works to extend the number of people involved with coding, publishing, building compilers.
- IEEE- similar
- IBM- they wanted to be on top of what was going on in the programming world, to know what was going on with their shares, they would also fund projects.
- IFIP- interest in machine and everything you could do with it.
- GAMM- German group of engineers, maths origin, working to get Germany economy up. they were promoting anything that was universal (door oorlog maybe)
- the rest were mainly pushing interests for their users groups (business maybe).

The ALGOL conspiracy/ history of ALGOL

- dream of a universal language. Darmstadt, Dresden, munched, Zurich, mainz. from there it went to the rest of Europe, amsterdam for example. but also east of europa. all of them like a universal language based on maths.
- it started to fall apart because of SHARE. if SHARE didn't mind, why would IBM.
- IFIP started a working group 2.1 for ALGOL. based on being able to add stuff to the original?
- ACM decided to make ALGOL the standard language of all scientific program. up until 1962 there was 1 only paper (?) so changing all of it wasn't that difficult, from there on everything needed to be ALGOL.

Rise of software industry

- software didn't have a good definition at first. code produced baby one of the companies programmers it was called code or program.
- beginning 60s software alone did not suffice to uphold a business: a range of computer service like maintenance, machine building/tuning and batch processing.
- software is soft?

IBM 1968: unbinding

- form that moment software becomes both soft and ware, idk for sure but maybe because it isn't something you can hold in your hands but you can sell it?
- academics in Europe had some problem with that idea (me too?)
- another thing went wrong: academics would benefit from a crisis?

software crisis

- problem with growing amount of code was the amount of punch cards.

AI and mathematics

- McCarthy 1961: computer programs for checking mathematica proofs- paper presented at the with AMS conference for pure mathematics.
- said the fields would work together as closely as analysis and physics
- in 1965 there was a proof checker for predicate calculus
- NG de bruin presented automath 1967, program to check proofs. he wanted it to correct proofs as well but that was too difficult.
- Automath project 1970-75 → Landau, grondlagen der analysis form 1930 checked completely.
- 4 colour theorem → being able to color maps such that no neighbouring countries have the same colour. it raised questions of 3-4 dimensions. they essentially brute forced it, going through all the colourings. Thomas Tymoczko says the computer didn't proof anything, he wanted more clarity.
- in 1970s automatic reasoning assistant (AURA) came. AURO could help you generate mathematical proof. Computer programming changed mathematics

Computers in education

- people in darthmouth started BASIC
- students had access to these
- computers were part of the curriculum they played a role in 3 parts...
- they introduced binary arithmetic (useless) en flowcharts. Eventually the language ECOLE came up for students to learn.
- learning to make exercises or learning to think → they need to understand understanding. solving equations doesn't necessarily involve thinking. they started making exercises that focus on thinking and understanding. they kept involving curriculums.
- in 1961 there is a comity formed CMLW to discuss what needed to be in the curriculum. in 1968 a report was formed that said it was a necessity to include computing within maths curriculum with as language ALGOL. the usual academic opinion was that they looked down on BASIC snd its usage.
- dutch schools started an extra course, computerkunde → programming is maths. some schools took Guus Vonk 1971 ideas and implemented them, they tried to make it connect to the real world, making match making algorithms. the course got discontinued fast.
- there were new requirements for schools/ new school systems so that pupils were ready for the future. Benjamin Bloom 1956 came up with system to track student progress, beneficial for teachers (meeting the need of teacher/ does teacher perform good).

thinking about education

- Grace Hopper, Skinner (machines) 1954 → programming means educating a computer (where AI comes in).
- programming books (algebra) in NL with fill in the blank question about the instructions. This was influenced by computing tech and businesses. there was a connection with AI in the sense that people making those books are thinking about thinking? what does it mean to solve a specific problem, they're rephrasing it in such a way that is it solvable by computers.
- PLATO 1960 was an example of computers solving problems for university students. Not all students had access though.
- 1970s came programming scrap cards for students? 1980 it was in classrooms?

6th of May

Lecture 5: Dream machines

Vince's Notes:

Personal Computing Evolution Overview

- Computing transformed from specialized tools to personal devices
- Shift from technical computing to accessible interfaces
- Windows and icons made computers user-friendly for general public

1980s Personal Computer Revolution

- Personal computers gained mainstream popularity in 1984
- Technology became accessible beyond technical specialists
- Movie theaters and NASA footage helped popularize computing concepts
- European adoption followed similar patterns to US market

Expert Systems Era (1970s-1980s)

- Software focused on rule-based expert systems
- Systematic approach to organizing knowledge and workflows
- Applications in:
  - Medical diagnosis and decision-making
  - Molecular structure analysis
  - Business decision processes
- Published systems appeared by 1986

Japanese Fifth Generation Computing Initiative

- Japan launched ambitious computing project in 1984
- Goal to create next-generation computers beyond transistor-based systems
- Previous generations:
  1. Tube-based computers
  2. Transistor-based systems
  3. Integrated circuits
  4. Microprocessors
- Expert systems and AI were central focus
- Created competitive pressure with US technology sector

Early AI and Game Development

- Tic-tac-toe programs demonstrated basic AI concepts
- Text-based adventure games popular in 1970s-1980s
- Simple question-response systems simulated intelligence
- Programs designed to appear “wise” through strategic responses
- Foundation for later AI development and user interaction design

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Aditi’s Notes:

Silicon Valley

- Ted Nelson, computer Lib (1974), Xanadu-project (1960)

Japanese valleys

- Part of “Silent History”
- He said something like the japanese look at machines as they are having souls

European Valleys

- European: academics and high tech companies were interested in time sharing facilities.
- Utopian city planning, for example : Sophia-Antipolis (france 1969). People went to school there, work, but no1 lived there, they lived in Nice a few miles down the coast. Until 1960s not a lot of ppl lived there, and gradually the city was crowded with ppl.
- They were just copying the idea of silicon valley from the US, there, also near Lyon
- They tried in the Netherlands there, but did not work, let’s build the valley there, a university, people will flock , companies will also, but it did not happen (it kind of does now) Twente Polytechnic (Enschede, 1964)
- It worked in Eindhoven “Valley”, ASML was there, Philips was there, later down it became the kind of valley atmosphere that they wanted in Enschede
- It might happen in the most random places, as in Mannheim(Germany), SAP issued R there in 1972 (can he finish an idea before starting another), bro came back to the idea and said they wanted to keep the engineers there, they built R, some super spreadsheet to help companies to streamline some things
- Main message of this kind is that outside the US there was some successful companies
- The youth were not interested in computers in Europe, the connection to how it was in the US(as hippies were interested in computers), was missing here
- In 1973 there was a Amateur computer club by Mike Lord. [They had a newspaper]
- Homebrew Computer Club in 1975
- Hobby Computer Club (HCC) 1977 in the Netherlands, so same acronym as the Homebrew Computer club
    - activities for members (there were no connection with the youth culture)
    - Exchanging knowledge and software
    - Programs in the newsletter
    - radio- broadcasting programs in NOS
    - no political agenda
    - Identity (as in they were not using a GUI, but command prompt line, the GUI was for …)
    - lack of political activism in the 60s and 70s
    - Rise of computer science in universities
    - There WAS continuity (cf Campbell-Kelly)

The rise and rise of expert systems

- Expert systems were part of AI, cybernetics systems
- Systems were not new, already in the 17th century the idea of systems started to rise
- Nature obeys to law, and the man cannot change the law, this belief gave rise to systems, said something about biological ones also, like divide the mammals, reptiles, whatever
- In the 20th century they not just use systems to describe the world, but as a way to control it, Jan Tinbergen (econometrics, 5 year plans(in the Communist countries, they were based on systems)
- RAND corporation tries to apply systems to warfare, IISA to diplomacy and library systems also for the knowledge or whatever
- All of this gives rise to expert systems as in software for companies
- Ideas were already there in 1956
    - In 1956 at namur conference Henryk Greniweski presented Logique et cybernetique (Cybernetics without mathematics)
    - Use cybernetics to control our factories
    - Stafford beer, the impact of cybernatics
- DENDRAL(1965(appearence), 1969(how to help you continue your research), 1973(became useful actually in 1973, before that was used by students mostly in help to continue research and to see what molecule you were looking at) in LISP, (large program to organise molecules or to help you figure out what molecule you were looking at) by Edward Feigenbaum, Bruce Bucchanan Joshua (and Esther) Lederberg, Carl Djerassi a.o.
- MYCIN(1970) in LISP
    - Something about bacteria infection
    - You put symptoms in the system, they told you what treatment to give, antibiotics or whatever
- IN USSR
    - Tamara Alexandridi(1967) build an expert systems to facilitate road building and build automobiles
    - Ekaterina Lavrischeva in GDR
- In China
    - Qian Xuesen (american citizen originally) he was sent back to China
    - He helped the chinese develop rocket plans, came up with all the ideas for cybernetics, organize the factories in China, and China
    - Mao praised him for organizing the electronic factories in such a way that even if they were invaded, they could help wager war or whatever
- We return back to the US, apparently they were presented before, I just missed them
    - Applied on local level in factories in USSR and US (did he say universities for the US, idk)
    - Dendral and MYCIN were in the US, bro just go and finish an idea
    - Software(and hardware) companies commercialising expert systems
        - 1980s: Lotus, Teknowledge, Intellicorp, Symbolics, IBM, Carnegie group, TI
- They extended cybernetics not only to biological systems, but to more companies
- Those companies hyped the ideas of expert systems, AAAI conferences were funded by these companies (can we say they had an agenda)
- Similar companies in the Europe, ACM(no, it stands for Association of computer machinery???)
- 1982 Japan launched the fifth generation initiative, hyped by IBM, 1st generation the dinosaurs with tubes and radio, 2nd generation built with transistors, 3rd generation was the transition to chips and processor, bro skipped 4th generation, 5th generation was smart computers
    - They were hijacking the IBM hype about computers and said we japanese will built this generation of smart computers#

- Pamela McCorduck and Feigenbaum, 1983 wrote the book the Fifth generation, saying the above, and that the US should follow
- 1980s still in the Cold war, and of course the americans said that the Japanese were beating them here and they were not taking it likely
- Building expert systems was there, 1989 textbook, building expert systems in prolog (european book)
- 1990s: demise of the expert system companies, systems themselves kept expanding
  - Japanese stopped pouring millions monthly on the 5th generation, waiting for better hardware
- Expert systems in the market place was not as useful as they hoped for AI and Games
- Start with the home computer
  - Commodore 64(1982) one of the early ones that had graphical capabilities(???)
  - The we have Commodore at home was launched in the same year Sinclair ZX spectrum(1982)
  - There was no GUI for both or just the latter, idk i zoned out for 3 seconds
  - The home computer was mostly used for playing games
- Playing games was also a way of appropriating
  - Some companies invented 1 million dollars to play tic tac toe
- Joseph Weizenbaum, in Datamation in 1962 started a newspaper which talked about how to make computer intelligent
  - Intelligence is hard to say to mimsky or this guy, but they can make it appear intelligent.
- Oregon Trail (1971) , MECC (1974) - AI. A text based adventure game. (They used it for history lessons in the US). Individuals had to give commands to the game and conduct basic communication with the computer program.
- Similar one ZORK 1 in PDP-10, based on game books from the 1930s
- Research into language recognition systems or automatic translating:
  - ELIZA(psycotherapist, pick up the main words and respond and try to keep you talking) - First language recognition system, by Joseph Weizenbaum 1964-1967
  - SHRDLU, Terry Winograd 1968-1970 (could respond and understand commands, like pick up the red block, if there were 2, it would ask you which one)

Rise of Game Culture

- Space war (1970) , One of the expensive machine hahahah
  - Created by MIT students (1961) on a DEC PDP-1 . Vectorbeam (1977) on a DEC PDP-architecture
- Pong 1972 (Atari became famous by the Pong arcade game)
- Pacman (1980), by Namco (1955): the Japanese take over. Adapting to PC culture. First on an arcade machine and then on software.
- Nintendo pushed atari out of the market, the great japanese market takeover (Donkey Kong (1981), by Nintendo (1889), featuring Mario)
- Strategy games, turn based (1991)
- Chess: the drosophila of AI
  - Drosophila: combination of organism, culture and practice
  - 1950 Claude Shannon: type A / type B Chess algorithm by Alan Turing (1952)
  - HAL in 2001 (1968) plays chess

Max Euwe (1901-1981) • UNIVAC ambassador, advisor in automation • Chess • 1964: professor in Automatic Dataprocessing (Rotterdam)

General approach: minimax-algorithm combined with an evaluation function and plausible moves 1997 Deep Blue vs Kasparov

11th of May

Lecture 6: Rise of Computer Culture

Transcription Notes:

Computer Ownership and Usage Statistics

- By 2000, 70% of Dutch people owned a personal computer
- Computer and internet use rose from 3.5 hours per week in 1995 to 3.9 hours per week, showing gradual adoption
- Computer ownership increased steadily from the 1980s onwards, though it remained a relatively narrow subculture through the 20th century

Computer Culture as Subculture in the 20th Century

- Computers were part of a subculture throughout most of the 20th century, with computer culture rising particularly in the 1990s
- The rise became more visible with smaller-scale technology developments
- Popular culture reflected this shift, with movies like *The Matrix* (1999) exploring themes of computer games and virtual reality
- Computer culture became more mainstream in the last couple of decades of the century, becoming visible and accessible to broader audiences

Hacker Culture and Freedom of Information Movement

- The freedom of information movement emphasized that computer networks should be free, accessible to everyone, and not controlled by corporate interests
- In the US, there was a hacker community, while in Europe different movements emerged, including the squatter movement in cities like Amsterdam, Hamburg, and Paris in the 1980s
- The Chaos Computer Club emerged from the squatter movement, advocating for network accessibility and freedom similar to the Legion of Doom in the US
- The Chaos Computer Club published a printed newsletter distributed through mail, promoting anarchist ideals around computing
- The movement claimed access to information networks as a new human right
- In 1986, the Chaos Computer Club hacked Deutsche Bank to demonstrate security vulnerabilities, showing that bank accounts were not secure
- This hack led to the establishment of the first laws against computer crimes and changed the relationship between banks and hacker groups
- The club gave talks across Europe and published information on topics like building modems, reviewed machines, shared software plans, and engaged in political activism
- The political activism missing in the 1970s manifested in the 1980s through anarchism and socialism in European hacker culture
- Various similar clubs existed across Europe, some public and some secret, all promoting the confluence of freedom of information

19th Century Telegraph Networks as Precursors

- Telegraph networks in the mid-19th century were built alongside railroad tracks to coordinate train schedules and prevent collisions

- Telegraph companies quickly realized the potential for sending messages between locations, leading to the rapid growth of commercial telegraph services
- Submarine cables were laid across ocean floors, connecting Boston, Central America, Europe, and England
- Between 1872 and 1922, the number of telegraph lines expanded dramatically, with many companies involved
- Companies needed to coordinate and establish protocols to enable communication across different networks
- This led to early international conferences and the formation of organizations that would later influence computer network protocols
- The technical organization that emerged from these telegraph network conferences laid groundwork that would later benefit computer networking

National Approaches to Network Development

Russia

- Local networks existed in Russia but were not well connected to each other
- Despite the centralized state economy that could have supported projects like Cybersyn, the political structure prevented open communications
- Different government departments built separate networks without interconnection

China

- In the 1990s, internet access in China was very limited and strictly controlled
- FidoNet was one popular service, and Chinese Spider was one of the most popular sites for learning about tech, computers, and networks
- Chinese Spider went down in the late 1990s

European Countries

- European countries had state-owned telecommunications companies that controlled network infrastructure
- Germany had Deutsche Bundespost (telecommunications), France had France Télécom
- These companies explored how to utilize their network capabilities

French Minitel System

- France Télécom developed Minitel to make their extensive telephone network more valuable and attract more subscriptions
- The French telephone network connected every village, as each village was supposed to have a telephone connection
- Every French village mayor had a phone, ensuring widespread infrastructure
- However, in rural areas in the 1970s, many people didn't need phones since they relied on village communication systems
- Minitel offered access to an information network with an internet-like feeling
- Users could access information about local restaurants, menus, and other services
- Minitel shut down in the late 1990s and early 2000s as people switched to internet service providers
- The question became: why subscribe to an internet provider when everything was already available through the telephone provider?
- This story is well described in Campbell's book

Dutch Network Infrastructure

- The Dutch situation was more complicated than France, with many players but no single big success
- PTT (the Dutch postal and telecommunications company) invested in videotext services similar to Minitel
- Television networks, run by foundations with government oversight, also had their own network infrastructure
- Television networks used teletext, borrowed from Britain, primarily to help hearing-impaired viewers
- Teletext was incorporated into most new television sets, and by the end of the 1980s most Dutch people had teletext capability
- Teletext was used occasionally, notably for quick access to football match results, but was mostly one-way traffic
- Dutch networks adopted the X.25 protocol as the most suitable standard for European networks
- The service was expensive, with subscription costs that monopolistic providers could set freely
- One-time connection costs ranged from 500 to 2,000 euros
- The network was intended for business use - journalists, corporations, banks - not home consumers
- From 1984 onwards, municipal archives and administrative services used these connections
- National traffic services and customs also used the network
- While subscription costs eventually dropped, the system struggled to gain widespread adoption
- Connecting existing networks made sense from an infrastructure perspective

Academic Networks and Their Role

- Academics viewed networks as extensions of their research environment, similar to having a library
- Universities wanted information from institutions like CERN in Amsterdam within minutes
- Academic networks used the TCP/IP protocol developed in the 1970s
- Academic groups had their own culture and sense of humor around networking
- NIKHEF (an Amsterdam-based European physics research facility) had major network infrastructure, including direct cables from the UK, US, and CERN
- Connections were often dial-up, used as needed, preferably during off-peak hours
- European university networks joined together in the EU-NET system
- Academic networks were influential despite lacking capital or formal authority

Hacker Magazines and Phone Phreaking

- Hacker publications from 1989 to 1994 provided information on phone tricks to avoid paying bills for long-distance calls
- These publications included disclaimers that some information might be illegal, but they provided it anyway
- Phone phreaking (manipulating phone systems) was a significant part of the culture
- The community used insider jokes and technical humor

Early Internet Infrastructure in the Netherlands

- Hackett was one of the groups involved in Bosch agriculture infrastructure in the 1990s
- They became the first Dutch internet provider, later absorbed by a larger network still existing today under the name "freedom of internet"
- Started with a small service using 20 dedicated telephone lines, representing a very small group among the first internet providers
- From 1989 onward, they organized annual hacker parties, which were among the first quantum creators

The Digital City Project (1993)

- One of the hacker parties resulted in huge funding from Amsterdam, allowing them to build a website for the Digital City
- The project made the city of Amsterdam available online in 1993
- Features included a post office with email and various navigational options
- The initiative was immensely popular, launched in 1993
- Public access points to the network were established, allowing people to do whatever they wanted
- This became a big influential player in the rise of the internet

Disappointment with Internet Commercialization

- The generation of hackers and academics involved in early internet development became the most disappointed about what the internet became
- When the web became commercial in the mid-1990s, many felt the internet gave away its promises
- Surveillance became the default rather than an option
- Data use became standard practice

Network Standardization and Stakeholders

- Standardization of networks involved many stakeholders and ideas
- In the Netherlands, many companies and institutes came together to make an open network
- Participants included network owners, companies, television companies, rail companies, and universities
- Big companies like Philips in Eindhoven had at least local networks in their offices
- Universities and related institutions were also involved, including CWI in Amsterdam and the nearby high energy physics lab
- Media companies were the first to have their websites and provide content as soon as it was commercially possible

Academic Research on Digital Culture

- References were made to books studying the rise of computer culture, including observations of children playing online
- Research examined how people can take on different characters depending on the game
- Second Life was mentioned as an example from the early years - a massive online world where people could buy clothes and houses
- Sherry Turkle's work was discussed, including observations about how social media changed public behavior
- A comparison was made showing people in the 1950s versus today - noting changes in attention and interaction
- Turkle observed that social media increased loneliness for some people
- Daniel Miller's book "How the World Changed Social Media" was mentioned, highlighting that people adopt social media for what they already needed
- Examples included Chinese students having two Facebook accounts - one for family and one for friends

History of Artificial Intelligence

- The first history of AI book was written in the 1970s and remains highly regarded
- The story begins with ancient Greek times, but reflection on thinking as something special and inherently human emerged with Descartes in the 1600s and 1700s
- The 1956 summer school was a key moment in AI history
- The first AI winter occurred, though it wasn't mentioned in some historical accounts

- Military funding was mentioned in relation to AI development
- Researchers didn't lose their university jobs or stop publishing during the AI winter

Information Systems and Search Engines

- Librarians were keen on using AI and cybernetics to make information accessible
- Until the 1970s, libraries used card catalogs in drawers sorted alphabetically and by topic
- Computer-based information systems emerged as alternatives to traditional card catalogs
- The hypertext idea came from these systems - click and hopefully find what you're looking for
- Search engines developed in the late 1990s and early 2000s
- Online advertising emerged with the idea of targeting based on what people are already looking for

Modern AI Development and Neural Networks

- The "new godfathers of AI" emerged in the 1990s
- Development involved statistics and distribution across layers
- A shift occurred from deterministic models to stochastic models
- Modern stochastics involves connecting distributions to several layers
- Key developments occurred around 2012-2016

AI Bias and Unheard Voices

- Concerns about AI bias and pattern recognition emerged before large language models
- Joy Buolamwini was confronted with facial recognition software that didn't recognize Black faces
- She worked on the "Algorithmic Justice" project to make companies and AI aware of biases in datasets and algorithms
- The concept of "unheard voices" is important to be aware of
- Historically, slaves were documented by traders but their own voices were not heard
- Today, low-end laborers are involved in training AI (such as for self-driving cars) but remain invisible
- An artist made a movie about these workers, showing how they've learned their work and would love to visit the places they're labeling but fear they won't be able to

Environmental and Ethical Concerns

- Timothy Bender and colleagues published "On the Dangers of Stochastic Parrots" in 2021 - the first academic paper with an emoji in the title
- The paper questions whether language models can be too big
- It focuses on environmental effects of computing
- Agbogbloshie in Accra, Ghana was mentioned as where computer waste ends up
- It used to be a thriving fishing community but there are no fish anymore because of pollution from discarded computers
- People living there are part of the unheard voices
- Large language models absorb significant energy during training and use
- The anthropomorphization of AI systems (like ChatGPT) was noted, recalling earlier observations about ELIZA
- While AI can construct beautiful texts and improve writing, it lacks true understanding